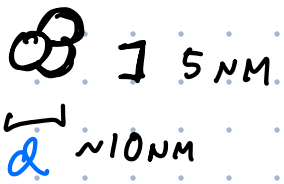
SCALES & ESTIMATES

1. MACHINE - MADE OF PROTEIN



- USES ENERGY FROM ATP $E \sim 100 pN\,nm$

GENERATE FORCE

$$F \sim \frac{E}{d} = \frac{100 pN\,\cancel{nm}}{10\,\cancel{nm}} \sim 10\,pN$$

2. NEWTON'S LAW

$$\sum F = ma \sim \frac{l}{t^2}$$

l ← DISTANCE, t ↑ TIME

ma → INERTIA

m - CELL
$l \sim 1\,\mu m$
$t \sim 1\,s$
$\Rightarrow ma \sim 10^{-4}\,pN$

$ma \ll \sum |F| \Rightarrow \sum F = 0$

ma INERTIA, $\sum |F|$ FORCES

NON-INERTIAL
LOW-REYNOLDS

3. DRAG



VISCOSITY $\eta = 10^{-3}$ Pa·s FOR WATER

$$= 10^{-3}\ \frac{N \cdot s}{m^2}$$

$$= 10^{-3}\ \frac{10^{12}\ pN\,s}{(10^{6}\ \mu m)^2}$$

$$= 10^{-3}\ \frac{pN\,s}{\mu m^2}$$

NUMBER

$$V = \left(\text{number}\right)\frac{F}{\eta d}$$

v, F, d, η



$$v \sim \frac{F}{\eta d}$$

$$\frac{\mu m}{s} \overset{?}{\sim} \frac{pN}{pN\,s}\,\frac{\mu m^2}{\mu m}$$

PLUGGING IN,

$$v \sim 10^2 - 10^3\ \frac{\mu m}{s}$$

FOR A SPHERE,

$$V = \frac{1}{6\pi}\,\frac{F}{\eta d}$$

HUMAN REFLEX

$d \sim 1.0\ m$

$t \sim 100\ ms$

$$v \sim \frac{d}{t} \sim 10\ \frac{m}{s}$$

4. GRAVITY IS NEGLIGIBLE

I. MECHANICS

SPRING



x - CONFIGURATION

$$E = E(x) = \frac{1}{2} k (x - x_0)^2$$

ENERGY

k, x_0 — PARAMETERS

GENERAL PRINCIPLE

$$F = -\frac{\partial E}{\partial x}$$

F — FORCE

E — ENERGY

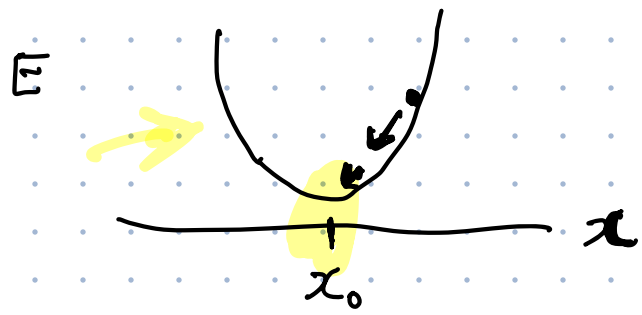
x — CONFIGURATION

$$\vec{F} = -\nabla E$$

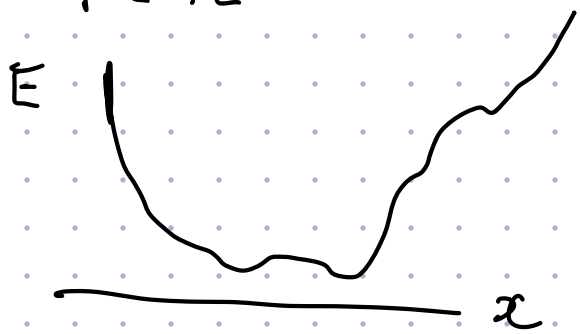
IN HIGHER DIMENSIONS

$$\Rightarrow F = -k(x - x_0)$$

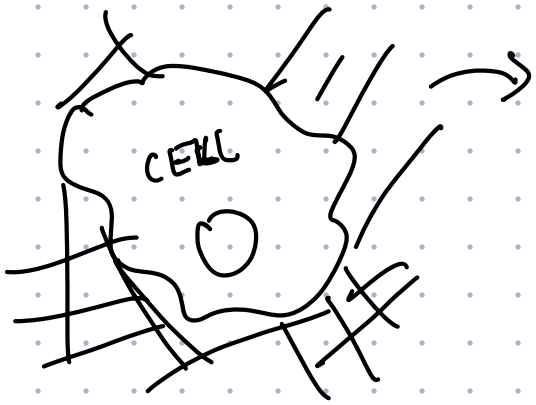
SPRING



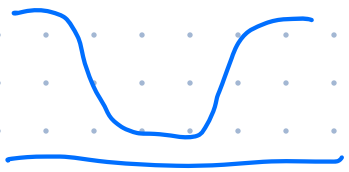
REAL ?



EXAMPLES



→ EXTRACELLULAR MATRIX

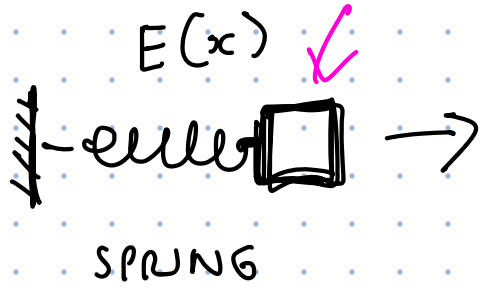


BME UCI

- W. LIU
- BOTVINICK

⇒ SYSTEMS EVOLVE TO MINIMIZE THEIR ENERGY

EXTERNAL FORCE



→ F_{EXT} EXTERNAL FORCE

REMINDER: IF $F_{EXT} = 0$, THEN

$$F = -\frac{\partial E}{\partial x}$$

WITH EXTERNAL FORCE

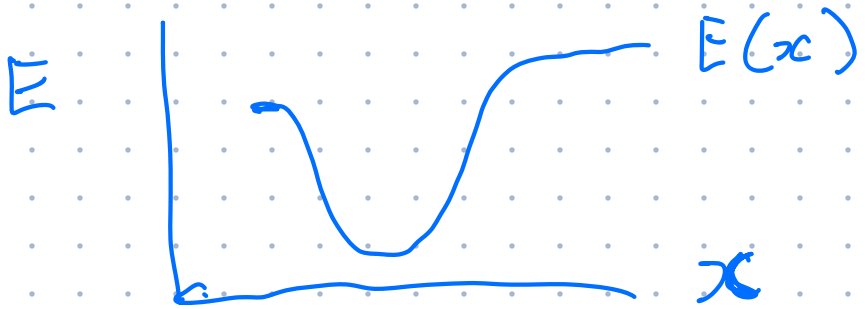


$$F = -\frac{\partial E}{\partial x}$$

PRETEND THAT $E_{FIC} = E(x) - x \cdot F_{EXT}$

↑ FICTITIOUS POTENTIAL

THEN $$-\frac{\partial E_{FIC}}{\partial x} = -\frac{\partial E}{\partial x} + F_{EXT}$$



$E_{FIC}(x)$?